# Lecture 3: Smoothed Analysis of Simplex via Circuits

## Algorithms & Geometry of Linear Programming

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# Worst-case instances are brittle

Klee–Minty (Lecture 2): exponentially many pivots — but brittle: small perturbations destroy the long path.

### The smoothed model [Spielman–Teng '04]

Adversary: $c$ and $(\bar{\mathbf{A}}, \bar{b})$, rows of norm $\leq 1$; nature: $\hat{\mathbf{A}}, \hat{b}$ with i.i.d. $N(0, \sigma^2)$ entries.

$$\max\ c^{\mathsf{T}} x \text{ s.t. } (\bar{\mathbf{A}} + \hat{\mathbf{A}})\, x \ \leq\ \bar{b} + \hat{b},$$

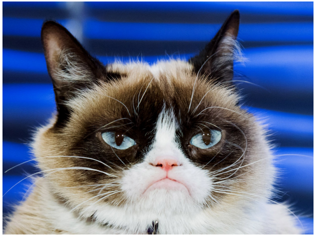
Bound the expected running time, worst case over the adversary.

### Theorem (Spielman–Teng '04)

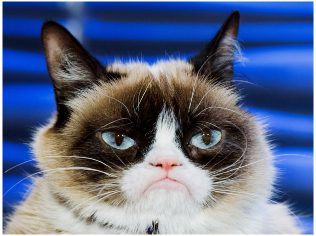
*The shadow vertex simplex method solves smoothed LPs with expected* $\mathrm{poly}(n, m, 1/\sigma)$ *pivots.*

Sharper bounds: [Deshpande–Spielman '05], [Vershynin '09], [D–Huiberts '18], [Huiberts–Lee–Zhang '23].

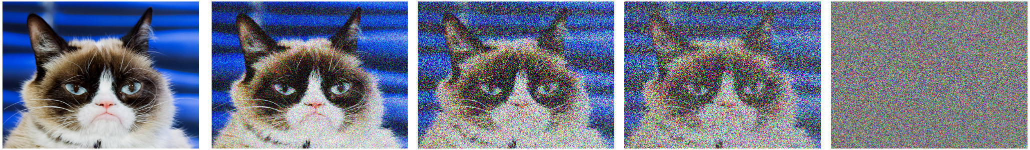
# Random is not typical



# Random is not typical




**Smoothed complexity:** [Spielman–Teng '04]



worst case, $\sigma = 0$

smoothed analysis, $\sigma$ variable

# Today: two complete smoothed analyses

Two complete analyses, perturbing only the objective:

1. **SSP** for minimum-cost flow
2. general LPs via **circuit imbalance**: constraint insertion

Common thread: circuits — simplex edge directions = flow cycles/paths.

# Outline

# The toolkit from Lecture 2

Standard form, $\mathbf{A} \in \mathbb{R}^{m \times n}$ of full row rank $m \leq n$:

$$\min c^{\mathsf{T}} x \text{ s.t. } \mathbf{A}x = b, \quad x \geq \mathbf{0}.$$

- **basis** $B \subseteq [n]$, $|B| = m$, $\mathbf{A}_B$ invertible; **BFS**: $x_B = \mathbf{A}_B^{-1} b \geq \mathbf{0}$, $x_N = \mathbf{0}$ — a vertex
- **pivot directions** at basis $B$, entering $i \notin B$:

$$w_i^{B,i} = 1, \qquad w_B^{B,i} = -\mathbf{A}_B^{-1}\mathbf{A}_i, \qquad w_j^{B,i} = 0 \text{ otherwise}$$

  — the edges at the vertex
- **variable bounds**: pair $x_j + \bar{x}_j = u_j$ — kernel notions carry over

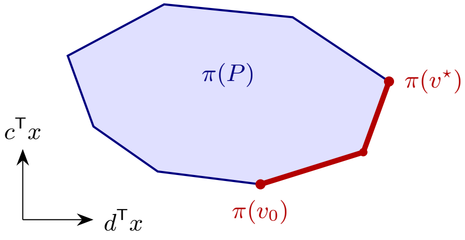
# The shadow vertex rule (recap)

*Shadow path for* $(c, d)$: start objective $c$ (minimizer $v_0$), target $-d$.

- maintain a vertex minimizing

$$c_\lambda := (1-\lambda)\, c - \lambda\, d, \qquad \lambda : 0 \to 1$$

- visited = lower boundary of the shadow $\pi(P)$, $\pi(x) = (d^\mathsf{T} x, c^\mathsf{T} x)$, walked rightward
- locality: visited $\iff$ minimizes *some* $c_\lambda$ — history-free




$v_0$ minimizes $c$, $v^\star$ maximizes $d$

# The local step: the tangent cone program

$T_P(v) = \{y : v + \varepsilon y \in P$ for small $\varepsilon > 0\}$; standard form: $\{y \in \ker(\mathbf{A}) : y_j \geq 0 \ \forall j : v_j = 0\}$.

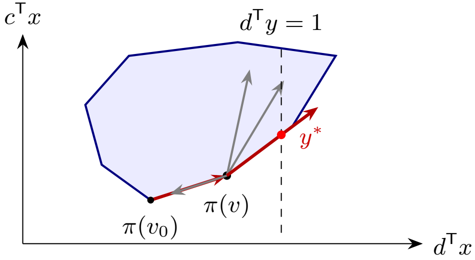
### Lemma (Tangent cone program)

*If $v$ on the shadow path minimizes $c_\lambda$ exactly for $\lambda \in [\lambda^-, \lambda^+]$, the edge followed out of $v$ solves*

$$\min \ c^\mathsf{T} y \text{ s.t. } y \in T_P(v), \ \ d^\mathsf{T} y = 1,$$

*with optimal value (slope)* $\dfrac{\lambda^+}{1 - \lambda^+}$.

slopes strictly increase along the path




lowest cone direction of unit $d$-progress

# Outline

Preliminaries

Smoothed Analysis of the Successive Shortest Path Algorithm

Circuit Decompositions & The Circuit Imbalance Measure

Smoothed Analysis via Circuit Imbalances

# Minimum-cost maximum flow

*This section:* $G = (V, E)$ directed, $n := |V|$, $m := |E|$, source $s$, sink $t$, capacities $u > \mathbf{0}$, costs $c \in (0,1)^E$.

- feasible flows:

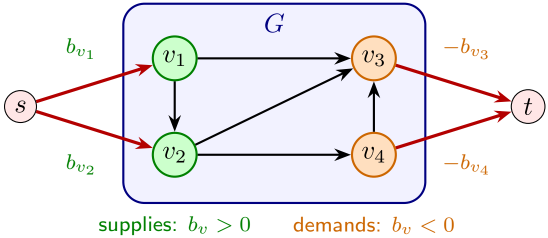
$$P := \left\{ f \in \mathbb{R}^E : \sum_{ij \in E} f_{ij} - \sum_{ji \in E} f_{ji} = 0 \;\; \forall i \neq s, t, \quad \mathbf{0} \leq f \leq u \right\}$$

- $\mathrm{val}(f) :=$ net flow leaving $s$
- incidence matrix $\mathbf{A}_G$: $\quad (\mathbf{A}_G f)_v =$ net outflow at $v$
- **goal**: min-cost flow among maximum-value flows

# Reduction from minimum-cost flow

Min-cost $b$-flow, for demands $b \in \mathbb{R}^V$ with $\sum_v b_v = 0$:

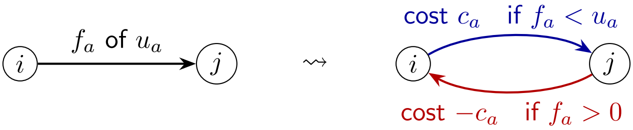
$$\min\ c^{\mathsf{T}} f \text{ s.t. } \mathbf{A}_G\, f = b, \quad \mathbf{0} \le f \le u.$$




- auxiliary arcs (red): cost $0$, capacity $|b_v|$
- max flows saturate all auxiliary arcs $\iff$ the instance is feasible

# The residual network and the SSP algorithm

For $f \in P$, the *residual network* $N[f]$ keeps, for each arc $a = ij$:



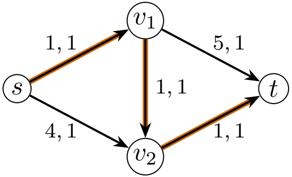

## Successive shortest paths (SSP)

Start with $f = \mathbf{0}$. While $N[f]$ contains an $s$–$t$ path: augment $f$ as much as possible along a minimum-cost $s$–$t$ path of $N[f]$. Return $f$.

Assume arising min-cost paths unique (probability $1$ after perturbation).

# SSP in action

iteration 1 $\quad (f = \mathbf{0})$

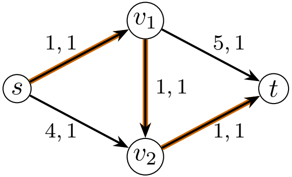



arcs: cost, capacity

cheapest path: cost $3$

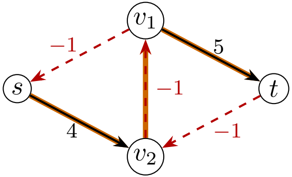
# SSP in action

iteration 1 $\quad (f = \mathbf{0})$




arcs: cost, capacity
cheapest path: cost $3$

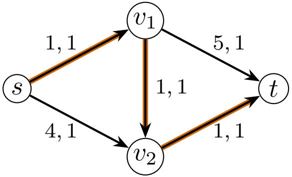
iteration 2: $\quad N[f_1]$




dashed: reverse arcs
cheapest path: $4 - 1 + 5 = 8$

# SSP in action

iteration 1 $\quad (f = \mathbf{0})$

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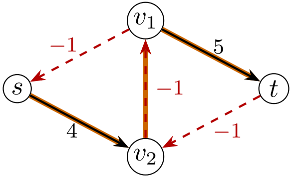
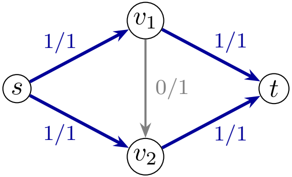



dashed: reverse arcs
cheapest path: $4 - 1 + 5 = 8$

final flow $\quad (f_a/u_a)$




no $s$–$t$ residual path:
min-cost max flow

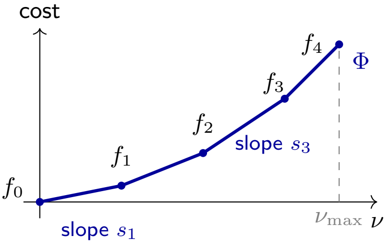
iteration 2 undoes the flow on $v_1 v_2$; $\quad$ slopes increase: $3 < 8$.

# SSP walks the cost curve

The *cost curve* $\Phi(\nu) := \min\{c^\mathsf{T} f : f \in P,\ \mathrm{val}(f) = \nu\}$ is convex, piecewise linear on $[0, \nu_{\max}]$.

### Theorem

*The SSP iterates $f_0 = \mathbf{0}, f_1, \dots, f_k$ are exactly the breakpoints of $\Phi$, in order, and the augmenting path costs $s_i$ are its slopes:* $0 < s_1 < \cdots < s_k < n - 1$.




- each $f_i$: min-cost of its value; last: min-cost max flow
- slopes: simple paths — $\leq n - 1$ arcs of cost in $(-1, 1)$

# SSP implements the shadow vertex method

The shadow path for $(c, \mathrm{val})$ walks the breakpoints of $\Phi$ — already justified. Remains: SSP performs its pivots.

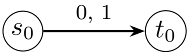
- the tangent cone program at a vertex $f$ on the curve:

$$\begin{aligned} \min\ c^{\mathsf{T}} y \text{ s.t. } \quad & (\mathbf{A}_G y)_v = 0 \ \ \forall v \neq s, t, \qquad \mathrm{val}(y) = 1, \\ & y_a \geq 0 \text{ if } f_a = 0, \qquad y_a \leq 0 \text{ if } f_a = u_a \end{aligned}$$

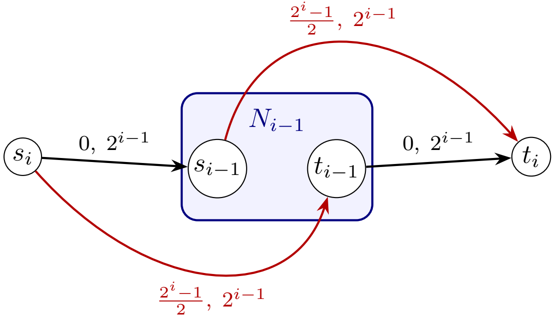
- sign constraints = residual arcs: $N[f]$ is the tangent cone, combinatorially
- optimum = cheapest $s$–$t$ path: one shortest path per pivot (exercise)
- [Zadeh '73; Disser–Skutella '19]: exponentially many iterations possible (next slide)

# Exponentially many iterations [Disser–Skutella '19]

Recursive counting network $N_i$ (arc labels: cost, capacity; their $v_i$'s set to $0$):




base case $N_0$




recursive step: $N_i$ from $N_{i-1}$, $i = 1, \dots, n$

- SSP on $N_n$: $2^n$ iterations, one unit of flow each
- the shortest $s_n$–$t_n$ path at iteration $j$ has cost exactly $j$, $j = 0, 1, \dots, 2^n - 1$

# The perturbation model, and the theorem

## Perturbation model

Each cost $c_a$, $a \in E$, is an independent continuous random variable on $(0, 1)$ with density at most $\phi \geq 1$. *Interval lemma*: $\Pr[c_a \in [\alpha, \beta]] \leq \phi\,(\beta - \alpha)$.

- example: uniform on an adversarial interval of length $1/\phi$
- $\phi \to \infty$: worst case — $\phi$ = adversary's precision

## Theorem (Smoothed SSP, Brunsch et al. '15)

*For any network with $n$ nodes and $m$ arcs, and random costs as above, the expected number of SSP iterations is $O(m\,n\,\phi)$.*

# Proof plan, and the charging step

**Challenge**: randomness not exhausted by the history. Locality: being on $\Phi$ is a property of the data.

- **arc sides** $(2m)$: $ij \in E \leftrightarrow f_{ij} \geq 0, \quad ji \in \overleftarrow{E} \leftrightarrow f_{ij} \leq u_{ij} \quad (c_{ji} := -c_{ij})$
- **charging**: iteration $i$ releases a tight bound — a side $a$ of its path; charge $i$ to $a$
- charged to $a \Rightarrow$ *the min-cost path uses* $a$:

$$p(f_{i-1}) \;=\; p^{+a}(f_{i-1}) \;=\; s_i$$

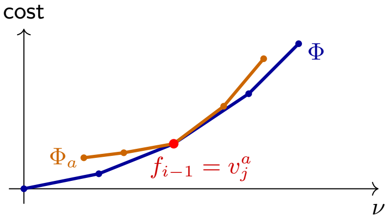
$$\underbrace{s \rightsquigarrow p}_{\text{indep. of } c_a} \xrightarrow[a]{\text{cost } c_a} \underbrace{q \rightsquigarrow t}_{\text{indep. of } c_a} \quad \Rightarrow \quad p^{+a}(f_{i-1}) \;=\; c_a + \big(\text{indep. of } c_a\big)$$

($p$: min path cost in $N[f]$; $p^{+a}$: min over paths using $a$.) Plan: expected charge per side $O(n\phi)$, from $c_a$ alone.

# The sandwich step

Facet $P_a := \{f \in P : f_a$ at its bound$\}$: curve $\Phi_a$, breakpoints $v^a_j$, slopes $s^a_j =$ $a$-avoiding path costs.

- charged to $a \Rightarrow f_{i-1}$ is a breakpoint $v^a_j$ of $\Phi_a$

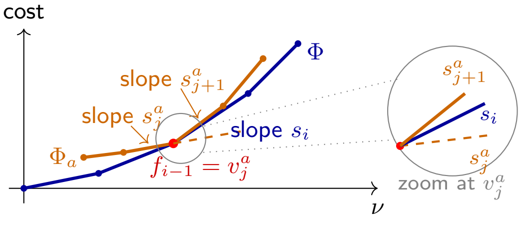


# The sandwich step

Facet $P_a := \{f \in P : f_a \text{ at its bound}\}$: curve $\Phi_a$, breakpoints $v_j^a$, slopes $s_j^a =$ $a$-avoiding path costs.

- charged to $a \Rightarrow f_{i-1}$ is a breakpoint $v_j^a$ of $\Phi_a$
- $\Phi \leq \Phi_a$, touching at $f_{i-1}$; clip at $0$ ($s_i > 0$):

$$\max(s_j^a, 0) \;\leq\; \underbrace{p^{+a}(v_j^a)}_{= s_i} \;\leq\; s_{j+1}^a$$




Distinct charges $\to$ distinct breakpoints:
#charged to $a \leq 2 + \sum_j \mathbf{1}[\max(s_j^a, 0) \leq p^{+a}(v_j^a) \leq s_{j+1}^a]$.

# The probabilistic step

Let $e \in E$ underlie the side $a = pq$ ($c_a = \pm c_e$); condition on all costs but $c_e$.

- **independence**: $\Phi_a$ — breakpoints *and* slopes — does not depend on $c_e$
- **unit coefficient**: paths through $a$ split as $(s–p) + (a) + (q–t)$:

$$p^{+a}(v_j^a) \;=\; c_a + r_j \;=\; \pm c_e + r_j, \qquad r_j \text{ independent of } c_e$$

- event $j$: $c_e$ in a fixed interval of length $\max(s_{j+1}^a, 0) - \max(s_j^a, 0)$
- **telescoping**: interval lengths sum to $\leq n-1$

$$\mathbb{E}[\#\text{iterations}] \;\leq\; 2m\,\bigl(2 + (n-1)\,\phi\bigr) \;=\; O(m\,n\,\phi). \qquad \square$$

# Where we are

**Block 1:**

- flow decomposition (proof still owed — next section)
- SSP $=$ shadow vertex method walking $\Phi$; one shortest path per pivot
- $O(mn\phi)$ from locality $+$ bounded slope budget

**Block 2:** general LPs — the circuit toolkit, then the algorithm.

# Outline

# What did the analysis actually use?

**Generic:**

- charging pivots to entering coordinates
- independence of the facet curve from the charged cost
- the nested-curves sandwich

**Flow-specific:**

- augmenting structures with entries in $\{0, \pm 1\}$: charged cost enters with unit coefficient
- path costs $\leq n - 1$: slopes in a bounded window

**Extension** to $\min c^{\mathsf{T}} x$, $\mathbf{A}x = b$, $x \geq \mathbf{0}$, smoothed $c$:

- unit coefficients $\to$ the circuit imbalance measure $\kappa(\mathbf{A})$
- bounded window $\to$ algorithm design: insert constraints *one at a time*

# Circuits of a subspace

### Definition (Circuit)

Let $W \subseteq \mathbb{R}^n$ be a linear subspace. A *circuit* of $W$ is a support-minimal nonzero $g \in W$: no nonzero $h \in W$ has $\operatorname{supp}(h) \subsetneq \operatorname{supp}(g)$.
Circuits of a *matrix* $\mathbf{A}$: the circuits of $\ker(\mathbf{A})$ — the minimal linear dependencies among its columns.

- depends only on the subspace ($\ker(\mathbf{A})$ here; $\operatorname{im}(\mathbf{A}^\mathsf{T})$ works too)
- **role 1**: circuits are the pivot directions of the simplex method (later this section)
- **role 2**: circuits are the elementary pieces into which every vector of $W$ decomposes without cancellation (next slide)

For incidence matrices: circuits $=$ *simple cycles* — two slides on.

# Conformal circuit decomposition

**Definition (Conformal vectors)**

$g$ *conforms* to $w$ if $\operatorname{supp}(g) \subseteq \operatorname{supp}(w)$ and $g_j w_j \geq 0$ for all $j$ (signs agree).

**Lemma (Conformal decomposition)**

*Every $w \in W \setminus \{\mathbf{0}\}$ is a sum $w = g^1 + \cdots + g^p$ of $p \leq \dim(W)$ circuits of $W$ conforming to $w$; cancellation-free: $|w_j| = \sum_i |g^i_j|$.*

Proof via the pointed cone

$$C_w := \{y \in W : y_j w_j \geq 0 \ \forall j \in \operatorname{supp}(w), \quad y_j = 0 \ \forall j \notin \operatorname{supp}(w)\} \ni w :$$

- **Minkowski**: $C_w = \operatorname{cone}(R(C_w))$
- **extreme rays are circuits**: the tight constraints of a 1D face cut out $\{y \in W : \operatorname{supp}(y) \subseteq \operatorname{supp}(g)\} = \operatorname{span}(g)$
- **Carathéodory**: keep independent generators — $p \leq \dim(W)$

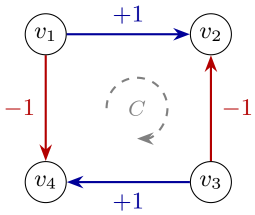
# Circulations and their circuits

Specializing to the incidence matrix $\mathbf{A}_G$:

- $\ker(\mathbf{A}_G) =$ circulations (negative entries $=$ flow against the arc)

**Lemma**

*The circuits of $\mathbf{A}_G$ are exactly the nonzero multiples of the signed indicators $\chi^C$ of simple cycles $C$.*




$\chi^C$: $+1$ on forward arcs, $-1$ on backward arcs

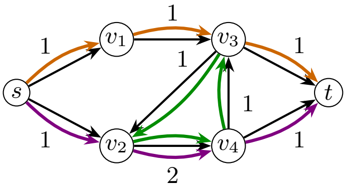
# Flow decomposition

### Theorem (Flow decomposition)

*Every $f \in P$ with $\mathrm{val}(f) = \nu \geq 0$ decomposes as*

$$f \;=\; \sum_i t_i\,\chi^{Q_i} + \sum_j t'_j\,\chi^{C_j}, \qquad t, t' > \mathbf{0}, \quad \textstyle\sum_i t_i = \nu,$$

*with $\leq m+1$ simple directed $s$–$t$ paths $Q_i$ and cycles $C_j$: all arcs in $\mathrm{supp}(f)$, traversed forwards. More generally, every circulation is a positive sum of $\leq m$ simple cycles conforming to it (forwards where $f_a > 0$, backwards where $f_a < 0$).*




$f \;=\; \chi^{Q_1} + \chi^{Q_2} + \chi^{C}$: two directed $s$–$t$ paths and one directed cycle, $\mathrm{val}(f) = 2$

# The circuit imbalance measure

**Definition (Circuit imbalance)**

For a linear subspace $W \subseteq \mathbb{R}^n$,

$$\kappa(W) \ := \ \max\left\{\frac{|g_j|}{|g_i|} \ : \ g \text{ a circuit of } W,\ i,j \in \operatorname{supp}(g)\right\},$$

the largest ratio between two entries of a circuit $\quad(\kappa := 1 \text{ if } W = \{\mathbf{0}\})$;
$\kappa(\mathbf{A}) := \kappa(\ker(\mathbf{A}))$.

- incidence matrices, and all totally unimodular $\mathbf{A}$: $\quad \kappa = 1$
- integral $\mathbf{A}$: $\quad \kappa \le \Delta_{\max}(\mathbf{A})$ (next slide)
- invariant under row operations; *not* under column scaling

# Circuits are the pivot directions

### Proposition

*Every pivot direction $w^{B,i}$ is a circuit, and every circuit of $\mathbf{A}$ is a scalar multiple of some $w^{B,i}$. Consequently*

$$\kappa(\mathbf{A}) \;=\; \max\left\{\left|(\mathbf{A}_B^{-1}\mathbf{A}_i)_j\right| \;:\; B \text{ basis},\; i \notin B,\; j \in B\right\}.$$

- $w^{B,i}$ support-minimal: smaller support $\Rightarrow$ supported on $B$ $\Rightarrow$ zero
- circuit $g$: extend $\operatorname{supp}(g) \setminus \{i\}$ to a basis $B$; then $g/g_i = w^{B,i}$
- used constantly: $\quad |\operatorname{supp}(g)| \le m+1, \quad \|g\|_1 \le (m+1)\,\kappa\,|g_\ell|$
- **Cramer**: $(\mathbf{A}_B^{-1}\mathbf{A}_i)_j = \pm\det(\mathbf{A}_{B-j+i})/\det(\mathbf{A}_B)$: $\quad \kappa =$ max determinant ratio of adjacent bases $\Rightarrow \kappa \le \Delta_{\max}$ for integral $\mathbf{A}$

# Stability of $\kappa$

**Lemma (Stability)**

*$\kappa(\mathbf{A})$ does not increase under:*

(i) *deleting a column;*

(ii) *contracting a column $i$: row-reduce it to some $e_j$, then delete row $j$ and column $i$;*

(iii) *duplicating a column up to sign: append $\pm\mathbf{A}_i$ as a new variable $y$.*

- (i): circuits of the smaller matrix are circuits of $\mathbf{A}$ avoiding column $i$
- (ii): on kernels, contraction *projects coordinate $i$ away*; a minimal-support lift of a circuit is a circuit of $\mathbf{A}$
- (iii): the only new circuit is the pair $(x_i, y)$, of ratio $1$

Exactly the operations relating the phase systems of the coming algorithm to $\mathbf{A}$.

# Outline

# Inserting one constraint at a time

**Assumptions:** $c_j$ independent, density $\le \phi$ on $(0,1]$; $\mathbf{A}_{[m]} = \mathbf{I}_m$ (row ops + permutation); truncated systems non-degenerate.

For $k \in [m]$, the *phase-$k$ trade-off curve* (convex, piecewise linear):

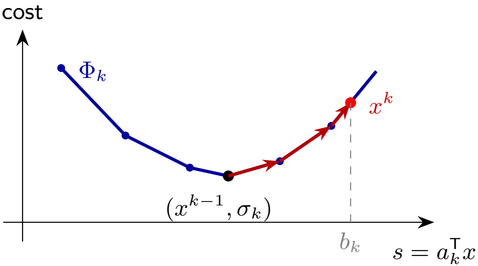
$$\Phi_k(s) \ := \ \min\Big\{c^\mathsf{T} x \ : \ \mathbf{A}_{[k-1],[n]}\, x = b_{[k-1]}, \quad a_k^\mathsf{T} x = s, \quad x \ge \mathbf{0}\Big\}, \qquad s \in I_k.$$

Constraint-insertion shadow simplex

Set $x^0 := \mathbf{0}$. For $k = 1, \ldots, m$: starting from $\big(x^{k-1}, \, \sigma_k := a_k^\mathsf{T} x^{k-1}\big)$, walk $\Phi_k$ monotonically from $s = \sigma_k$ to $s = b_k$; if $b_k \notin I_k$, stop and report *infeasible*. Return $x^m$.

# The phases chain together

- **idle variables rest**: on $\Phi_k$, $x_j = 0$ for $j > k$
- **chaining**: $(x^{k-1}, \sigma_k) =$ global minimum of $\Phi_k$ $\quad$ ($s$ is free)
- each phase starts at the bottom, climbs to level $b_k$




Complete solver: $b_k \notin I_k \Rightarrow$ infeasible; no Phase 0 needed.

# The phase systems inherit $\kappa$

In variables $(x, s)$, phase $k$ has constraint matrix

$$\mathbf{M}_k := \left[\, \mathbf{A}_{[k],[n]} \;\middle|\; -e_k \,\right] \qquad \big(\mathbf{A}_{[k-1]}x = b_{[k-1]},\;\; a_k^\top x - s = 0\big).$$

- **design point**: $-e_k$ duplicates a present column; the interpolation functional is the *coordinate* $s$
- "$d^\top w$" is an *entry of the circuit*: no cancellations

Lemma

*$\kappa(\mathbf{M}_k) \le \kappa(\mathbf{A})$ for every $k$, and every circuit of $\mathbf{M}_k$ has support of size at most $k+1$.*

# The smoothed guarantee

**Theorem (Smoothed constraint insertion)**

*Phase $k$ performs at most $n\big(2 + (k+1)\kappa^2\phi\big)$ pivots in expectation; over all $m$ phases,*

$$\mathbb{E}[\#\textit{pivots}] \;\le\; 2nm + 2nm^2\kappa^2\phi \;=\; O\big(n\,m^2\,\kappa^2\,\phi\big).$$

The SSP proof applies *word for word* — the dictionary:

| **SSP** | **phase $k$ of constraint insertion** |
|---|---|
| iteration | pivot along the curve $\Phi_k$ |
| arc-side $a$ (at most $2m$) | entering coordinate $a \in [n]$ |
| facet curve $\Phi_a$ (paths avoiding $a$) | curve $\Phi_{k,a}$ (column $a$ deleted) |
| clipped slope window $n-1$ | clipped slope window $(k+1)\kappa$ |
| $p^{+a} = \pm c_a + r_j$ (unit coefficient) | slope derivative $w_a/w_s \in [1/\kappa, \kappa]$ |

# The dictionary, in detail

- **slopes**: each edge is a circuit $(w, w_s)$ of $\mathbf{M}_k$:

$$\left|c^{\mathsf{T}} w / w_s\right| \;\le\; \|c\|_\infty \, \|w\|_1 \,/ |w_s| \;\le\; (k+1)\,\kappa$$

- **charging**: to the entering coordinate $a$ (bound $x_a = 0$ released)
- **sandwich**: $\Phi_{k,a}$ (column $a$ deleted) involves neither $x_a$ nor $c_a$
- **sweep**: entering-$a$ slope affine in $c_a$, derivative $w_a/w_s \in [1/\kappa, \kappa] \Rightarrow$ one interval of length $\le \kappa \cdot$ gap

Telescope over the clipped window $(k+1)\kappa$: expected charge $\le 2 + (k+1)\kappa^2\phi$ per coordinate. $\square$

# Discussion

- $\kappa = 1$: bound $O(nm^2\phi)$ — a factor $m$ above SSP; amortizing the phases is open
- why a fresh coordinate: a fixed $d$ can pair *degenerately* with circuits ($d^\mathsf{T} w \approx 0$); a coordinate cannot
- alternative: randomize the interpolation objectives — works for worst-case $c$ [Brunsch–Röglin '13; Dadush–Hähnle '16]

# Summary and what comes next

**Today:**

- *circuits*: pivot directions $=$ flow cycles/paths; the measure $\kappa$
- *conformal decomposition* (Minkowski $+$ Carathéodory) $\Rightarrow$ flow decomposition
- SSP: $O(mn\phi)$; constraint insertion: $O(nm^2\kappa^2\phi)$ — *locality* $+$ *bounded slope budget*

**Next lecture:** interior point methods — unconditional guarantees; circuits reappear.

*Exercises: SSP implements the shadow vertex method; source and sink arcs; flow decomposition from conformal decomposition; self-duality of $\kappa$.*